

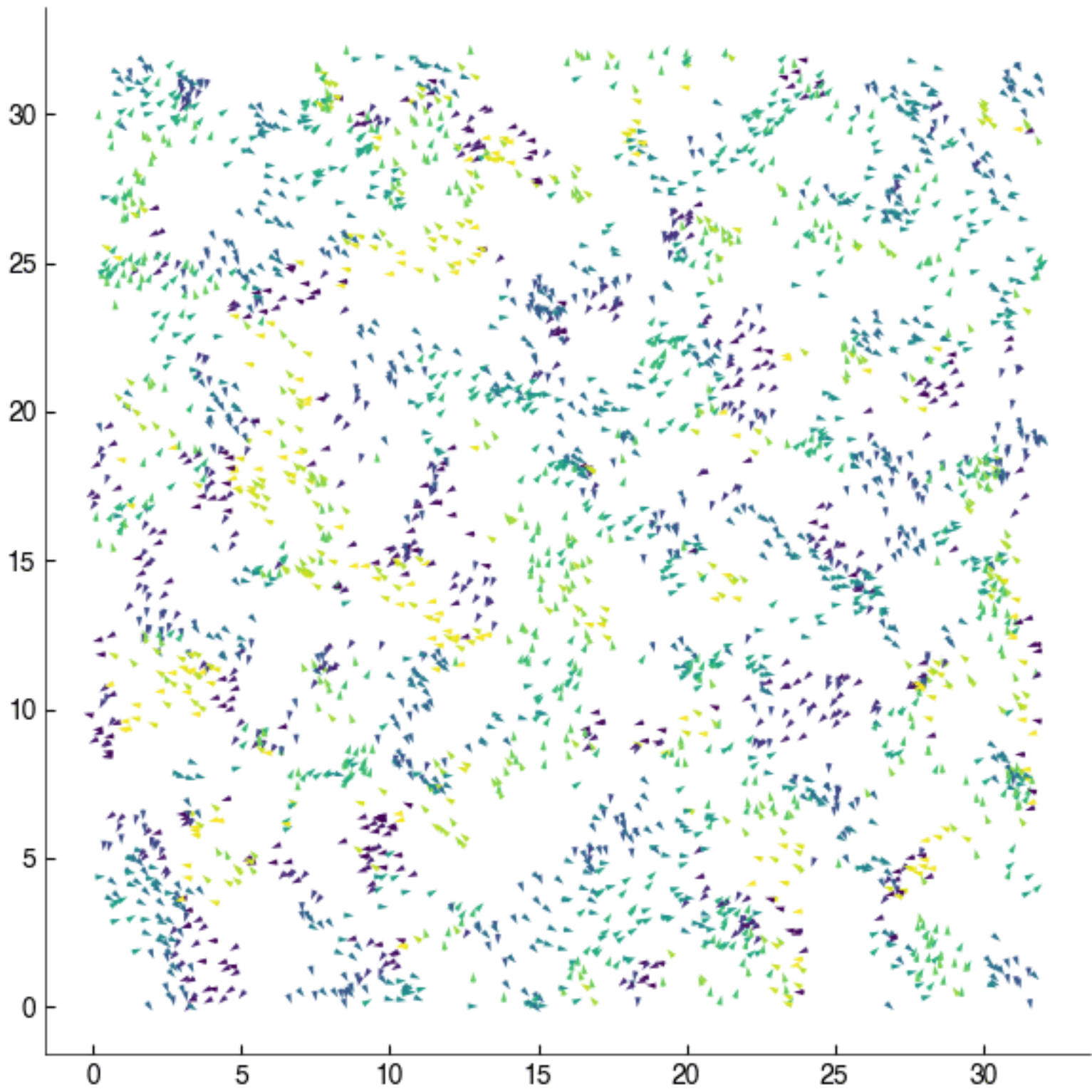
$$\begin{cases} \mathrm{d}X_t^i &= \mathrm{Pev}(\theta_t^i)\mathrm{d}t + \sqrt{2\sigma_x}\mathrm{d}W_t^i \\ \mathrm{d}\theta_t^i &= \chi F\left(\left(X_t^i, \theta_t^i\right), \mu_t^N\right)\mathrm{d}t + \sqrt{2}\mathrm{d}B_t^i \end{cases}$$

- Interaction F with strength $\chi > 0$

• Famous interacting active particle model:

- Vicsek model (Phys. Rev. Lett., 1995) for flocking birds, 6000+ citations

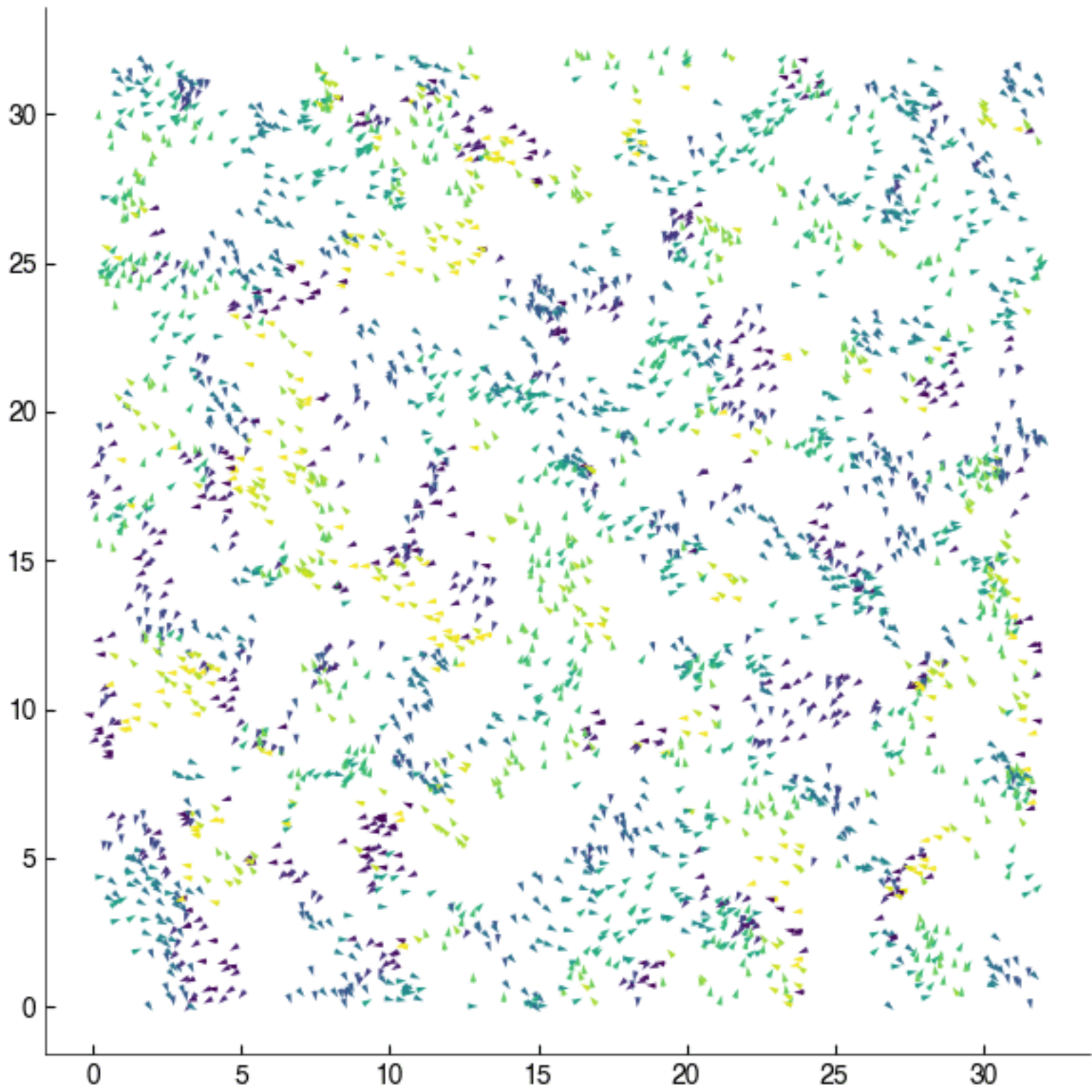
Interacting ants





F.

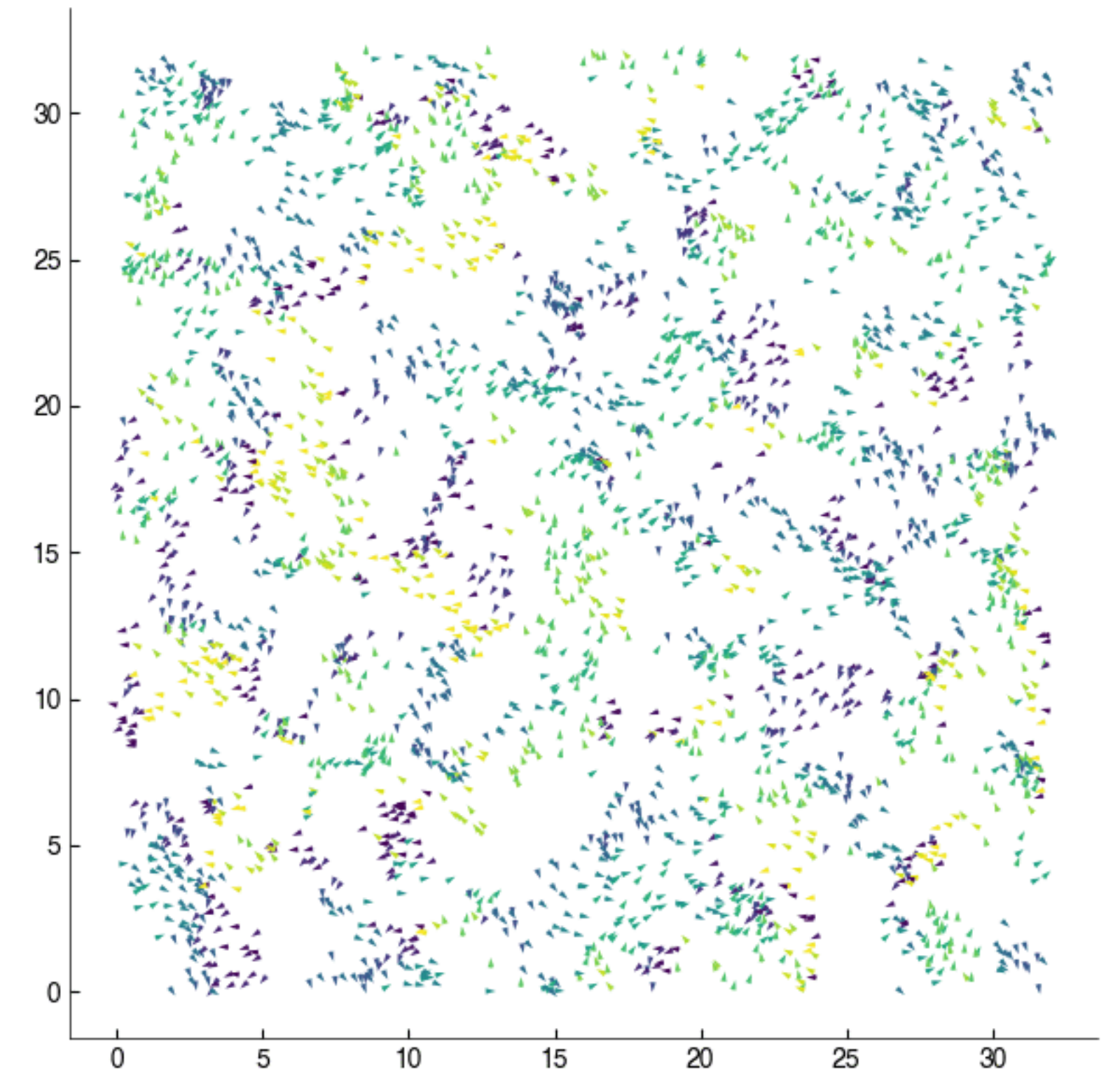
Turci



Interacting ants

$$\begin{cases} dX_t^i &= \text{Pev}(\theta_t^i)dt + \sqrt{2\sigma_x}dW_t^i \\ d\theta_t^i &= \chi F\left((X_t^i, \theta_t^i), \mu_t^N\right)dt + \sqrt{2}dB_t^i \end{cases}$$

- Interaction F with strength $\chi > 0$
- Famous interacting active particle model:
 - Vicsek model (Phys. Rev. Lett., 1995) for flocking birds, 6000+ citations



Interacting ants

$$\begin{cases} dX_t^i &= \text{Pev}(\theta_t^i)dt + \sqrt{2\sigma_x}dW_t^i \\ d\theta_t^i &= \chi F\left((X_t^i, \theta_t^i), \mu_t^N\right) dt + \sqrt{2}dB_t^i \end{cases}$$

- Ant model (dW, Bruna & Burger, SIAM J. Appl. Dyn. Syst., 2025):
$$F = n(\theta_t^i) \cdot \frac{1}{N} \sum_{j \neq i} \nabla K(X_t^i + \tau v(\theta_t^i) - X_t^j)$$